

[1] STARTUP SEQUENCE

1. Connect Master Phone A (via USB Tethering) to Laptop A[cite: 2].
2. Connect RJ45 LAN/Switch to Registration Laptops[cite: 2].
3. Turn ON Mobile Hotspot on Laptop A[cite: 2].
4. Connect Mobile Scanners (up to 8) to Laptop A's Hotspot[cite: 2].
5. Launch server_hub.py -> Click "► Start Engine"[cite: 2].
6. Start Cloudflare Tunnel (if online) -> Share public link[cite: 2].

[2] DEVICE ROUTING & URL MATRIX (DUAL-ENGINE ARCHITECTURE)[cite: 2]

DEVICE	NETWORK	TARGET URL (ENGINE)	PURPOSE
Kiosk Laptops	LAN	http://<IP>:5000 (Waitress)	Fast data entry (No SSL lag)[cite: 2].
Mobile Scanners	Wi-Fi	https://<IP>:5001 (Cheroot)	Unlocks iOS/Android cameras[cite: 2].
Master Phone A	USB Tether	https://<IP>:5001 (Cheroot)	Hardwired scanner; immune to lag[cite: 2].
Gate Displays	Wi-Fi/LAN	https://<IP>:5001 (Cheroot)	Instant GUI updates (Unbuffered)[cite: 2].
Roving Staff	4G/5G	https://<tunnel>.trycloudflare	Secure remote scanning[cite: 2].
Background Sync	USB/Wi-Fi	(Runs Automatically)	Cloudinary Photo API[cite: 2].

[3] GOLDEN RULES & REMINDERS

- STARTUP ORDER: Network -> Hotspot -> Server -> Clients[cite: 2].
- REGISTRATION PCs = HTTP (Port 5000)[cite: 2].
- SCANNERS & GUI = HTTPS (Port 5001) -> STRICTLY REQUIRED for cameras/live-sync![cite: 2]
- NEVER close server_hub.py during the live event[cite: 2].
- Staff never connect to MySQL directly; APIs handle all traffic[cite: 2].

[4] OFFLINE MODE (INTERNET GOES DOWN)

- DO NOT PANIC: Local Hotspot & LAN continue working normally[cite: 2].
- Registration uses Port 5000; Scanners use Port 5001[cite: 2].
- Remote Cloudflare staff will be DOWN until internet returns[cite: 2].

[5] SYSTEM & POWER OPTIMIZATIONS (CRITICAL)

- NETWORK PROFILE: Must be "Private" (Windows blocks LAN on "Public")[cite: 2].
- FIREWALL PORTS: Allow TCP 5000, 5001, 3306[cite: 2].
- ANTIVIRUS: Pause McAfee/Norton/Avast firewalls (they block local traffic)[cite: 2].
- POWER PLAN: Set Screen & Sleep to "Never"[cite: 2].
- ADAPTERS: Device Manager > Network Adapters > Uncheck "Allow computer to turn off device"[cite: 2].

[6] TROUBLESHOOTING & FIXES

» FIX: DEVICES CAN'T LOAD THE HUB IP

1. Check Laptop A's IP via 'ipconfig'[cite: 2].
2. Verify devices are on the correct Wi-Fi/LAN[cite: 2].
3. Turn OFF all VPNs (they scramble local routing)[cite: 2].

» FIX: CLOUDFLARE WON'T OPEN

- Change Laptop A DNS to: 1.1.1.1 (Preferred) and 1.0.0.1 (Alternate)[cite: 2].

» FIX: NETWORK GLITCHY / DROPPING

- Admin PowerShell Command (Run to force allow ports)[cite: 2]:
New-NetFirewallRule -DisplayName "EventHub Ports" -Direction Inbound -Action Allow -Protocol TCP -LocalPort 5000,5001
- Admin CMD (Network Reset)[cite: 2]:
ipconfig /flushdns
ipconfig /release
ipconfig /renew
netsh winsock reset (Requires Restart)

» MANUAL LAN SETUP (FALLBACK)[cite: 2]

- Server IP: 192.168.10.1 (Subnet: 255.255.255.0)
- Client IPs: 192.168.10.2+ (Subnet: 255.255.255.0)
- Client URL: http://192.168.10.1:5000

[7] CERTIFICATE INSTALLATION (HTTPS WARNING FIX) - UPDATED ARCHITECTURE

» WINDOWS PC INSTALLATION:

1. Locate 'rootCA.pem' (found in config/certs) and rename to 'rootCA.crt'.
2. Double-click the file and click 'Install Certificate'.
3. Select 'Local Machine' as the Store Location.
4. CRITICAL: Choose 'Place all certificates in the following store' -> Browse -> 'Trusted Root Certification Authorities'.
5. Click Finish and completely restart your web browser.

» ANDROID / iOS DEVICE INSTALLATION:

1. Transfer the 'rootCA.pem' certificate to the phone and rename it to 'rootCA.cer'.
2. WARNING: Do NOT open the file directly from the File Manager.
3. Open Android Settings -> Security & Privacy -> More security settings -> Encryption & credentials.
4. Tap 'Install a certificate' -> MUST select 'CA Certificate' (Do not select User/VPN).
5. Accept the privacy warning ('Install anyway') and select your 'rootCA.cer' file.
(For iOS: Install Profile in Settings -> General -> About -> Certificate Trust Settings -> Enable Full Trust).

[8] MYSQL DATABASE & NETWORK TUNING (LAN & PERFORMANCE)

» 1. ENABLE LAN / RJ45 / WI-FI ACCESS (my.ini REQUIRED)

- CRITICAL: Network binding cannot be changed live. It MUST be done in the file[cite: 2].
- Open: C:\ProgramData\MySQL\MySQL Server 8.4\my.ini (Note: ProgramData is a hidden folder)[cite: 2].
- Find the [mysqld] section and add/change this exact line (Use * to support IPv4 & IPv6 natively)[cite: 2]:
bind-address=*
- Save the file and restart the MySQL Service (Admin CMD: net stop MySQL84 && net start MySQL84)[cite: 2].

- » 2. ADD REMOTE USER FOR LAN ACCESS (MySQL Shell)
 - Open Command Prompt and log in: mysql -u root -p[cite: 2]
 - Run these exact commands to allow any laptop/PC on the network to connect[cite: 2]:


```
CREATE USER IF NOT EXISTS 'event_admin'@'%' IDENTIFIED BY 'EventHub123!';
GRANT ALL PRIVILEGES ON *.* TO 'event_admin'@'%';
FLUSH PRIVILEGES;
EXIT;
```
 - Other laptops can now connect using the Host Laptop's IP (e.g., 192.168.1.X) and these credentials[cite: 2].

- » 3. PERFORMANCE TUNING: METHOD A - PERMANENT TEXT EDIT (my.ini)
 - Open my.ini and locate the [mysqld] section[cite: 2].
 - Paste the complete Event Hub tuning block directly into the file[cite: 2]:

```
# === EVENT HUB - PERFORMANCE TUNING (4GB RAM BASELINE) ===
max_connections=150
max_connect_errors=10000
wait_timeout=600
interactive_timeout=600
net_read_timeout=60
net_write_timeout=120
max_allowed_packet=64M
tmp_table_size=32M
max_heap_table_size=32M
table_open_cache=2000
thread_cache_size=50
sort_buffer_size=1M
join_buffer_size=1M
read_buffer_size=256K
read_rnd_buffer_size=256K
default-storage-engine=InnoDB
innodb_file_per_table=1
innodb_flush_method=unbuffered
innodb_flush_log_at_trx_commit=2
innodb_log_buffer_size=16M
innodb_redo_log_capacity=256M
innodb_io_capacity=500
innodb_io_capacity_max=1000
disable-log-bin
slow_query_log=1
long_query_time=2
character-set-server=utf8mb4
collation-server=utf8mb4_unicode_ci

# --- RAM BASED SETTINGS (Set based on Host PC Total RAM) ---
# [ 4GB RAM -> 1G ] [ 8GB RAM -> 4G ] [ 16GB RAM -> 8G ] [ 24GB RAM -> 12G ] [ 32GB RAM -> 16G ]
innodb_buffer_pool_size=1G
innodb_buffer_pool_instances=1
# =====
```

- Requires restarting the MySQL service (Admin CMD: net stop MySQL84 && net start MySQL84)[cite: 2].

- » 4. PERFORMANCE TUNING: METHOD B - LIVE SHELL (NO MY.INI EDITING)
 - Use this to apply changes INSTANTLY without restarting the server[cite: 2].
 - Using 'SET PERSIST' saves the changes for future reboots without touching my.ini[cite: 2].
 - Note: 'innodb_log_buffer_size', 'innodb_buffer_pool_instances', 'innodb_flush_method', and 'disable-log-bin' MUST be done via Method A[cite: 2].
 - Open MySQL Shell (mysql -u root -p) and paste this block[cite: 2]:

```
SET PERSIST max_connections = 150;
SET PERSIST max_connect_errors = 10000;
SET PERSIST wait_timeout = 600;
SET PERSIST interactive_timeout = 600;
SET PERSIST net_read_timeout = 60;
SET PERSIST net_write_timeout = 120;
SET PERSIST max_allowed_packet = 67108864;
SET PERSIST tmp_table_size = 33554432;
SET PERSIST max_heap_table_size = 33554432;
SET PERSIST table_open_cache = 2000;
SET PERSIST thread_cache_size = 50;
SET PERSIST sort_buffer_size = 1048576;
SET PERSIST join_buffer_size = 1048576;
SET PERSIST read_buffer_size = 262144;
SET PERSIST read_rnd_buffer_size = 262144;
SET PERSIST innodb_file_per_table = 1;
SET PERSIST innodb_flush_log_at_trx_commit = 2;
SET PERSIST innodb_redo_log_capacity = 268435456;
SET PERSIST innodb_io_capacity = 500;
SET PERSIST innodb_io_capacity_max = 1000;
SET PERSIST slow_query_log = 1;
SET PERSIST long_query_time = 2;
SET PERSIST character_set_server = 'utf8mb4';
SET PERSIST collation_server = 'utf8mb4_unicode_ci';
```

- Set Buffer Pool dynamically (values MUST be in exact bytes in the shell)[cite: 2]:


```
[ 4 GB RAM PC -> 1G ] : SET PERSIST innodb_buffer_pool_size = 1073741824;
[ 8 GB RAM PC -> 4G ] : SET PERSIST innodb_buffer_pool_size = 4294967296;
[ 16 GB RAM PC -> 8G ] : SET PERSIST innodb_buffer_pool_size = 8589934592;
[ 24 GB RAM PC -> 12G ] : SET PERSIST innodb_buffer_pool_size = 12884901888;
[ 32 GB RAM PC -> 16G ] : SET PERSIST innodb_buffer_pool_size = 17179869184;
```

- » 5. VERIFY CHANGES (MySQL Shell)
 - To confirm LAN is open and RAM is allocated, open MySQL Shell and run[cite: 2]:


```
SHOW GLOBAL VARIABLES WHERE Variable_name IN ('bind_address', 'innodb_buffer_pool_size', 'max_connections');
```